Topic 05: Properties of Laplace Transforms

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

What if you could find $\mathcal{L}\{e^{3t}\sin(2t)\}$ without touching a single integral? You already know $\mathcal{L}\{\sin(2t)\} = 2/(s^2+4)$ from the standard table. The First Shifting Theorem says: multiply $f(t)$ by e^{at} in the time domain $\to$ shift s to $s-a$ in the transform. So $\mathcal{L}\{e^{3t}\sin(2t)\} = 2/((s-3)^2+4)$. Done. No integration. Just a substitution. This topic gives you five such shortcuts — once learned, they eliminate hours of integral computation.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this. The standard Laplace transform table (Topic 04) covers only "pure" time-domain functions: t^n, e^{at}, $\sin(bt)$, $\cos(bt)$, and so on. Real engineering signals, however, are rarely that clean. A damped oscillation is $e^{-2t}\sin(5t)$; a delayed ramp is $(t-3)\,u(t-3)$; a scaled signal is $\sin(3t)$. Without operational properties, computing any of these Laplace transforms means restarting from the definition integral every single time — which is tedious and error-prone.

The five properties introduced in this topic act as *transform algebra rules*. Each one takes a known result from the table and lets you convert it into the answer for a more complicated function — in seconds, not minutes.

Mistake	**Consequence**
Ignoring the First Shifting Theorem	Spending 10 minutes on an integral that takes 10 seconds with the theorem
Confusing s-shifting with t-shifting	Applying the wrong theorem and getting completely wrong answers
Forgetting $u(t-a)$ in the Second Shifting Theorem	Missing the unit-step multiplier in the inverse transform answer

Key Takeaway

Every property in this topic is a time-saver. Learn the statement, understand the proof once, then use it as a lookup rule. Properties are the reason the Laplace transform is useful in engineering practice.

3 You Already Know This (Intuition First)

Think of a corporate phone system. The main office toll-free number is 1800-XXX-XXXX — that is $F(s)$, the known transform. But every department has an extension: press +3 and you reach the Engineering Department, +5 and you reach Finance. Dialling the extension is exactly what the First Shifting Theorem does: it adds a shift to s, routing you to a "nearby" function in the transform domain.

Similarly, when you record a voice message and play it back 2 seconds later, you have introduced a *time delay* of 2. The Second Shifting Theorem captures this idea: delaying a signal by a units in the time domain multiplies its Laplace transform by e^{-as} — a clean, predictable scaling in the s-domain.

The Change of Scale property is like playing a video at $2\times$ speed or $0.5\times$ speed: you compress or stretch time, and the transform rescales accordingly. You do not need to re-derive anything; the rule gives you the answer directly from what you already know.

4 Property 1 — First Shifting Theorem (s-Shifting)

Theorem: First Shifting Theorem

Statement. If $\mathcal{L}\{f(t)\} = F(s)$ for $s > s_0$, then

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a), \quad s > s_0 + a.$$

Proof.

$$\begin{aligned}
\mathcal{L}\{e^{at} f(t)\} &= \int_0^\infty e^{-st}\, e^{at}\, f(t)\, dt \\
&= \int_0^\infty e^{-(s-a)t}\, f(t)\, dt \qquad \text{(combine exponents)} \\
&= F(s-a). \qquad \text{(definition of } \mathcal{L}\{f(t)\} \text{ with } s \text{ replaced by } s-a)
\end{aligned}$$

Inverse form. $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}\, f(t)$.

Worked Application: Find $\mathcal{L}\{e^{3t}\sin(2t)\}$.
Step 1. Identify $f(t) = \sin(2t)$, $a = 3$.
Step 2. From the standard table: $\mathcal{L}\{\sin(2t)\} = \dfrac{2}{s^2+4} = F(s)$.
Step 3. Apply the First Shifting Theorem — replace s by $s-3$:

$$\mathcal{L}\{e^{3t}\sin(2t)\} = F(s-3) = \frac{2}{(s-3)^2+4}.$$

Quick reference — First Shifting Theorem applications:

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$
1	$\frac{1}{s}$	$\frac{1}{s-a}$
t^n	$\frac{n!}{s^{n+1}}$	$\frac{n!}{(s-a)^{n+1}}$
$\sin(bt)$	$\frac{b}{s^2+b^2}$	$\frac{b}{(s-a)^2+b^2}$
$\cos(bt)$	$\frac{s}{s^2+b^2}$	$\frac{s-a}{(s-a)^2+b^2}$
t^2	$\frac{2}{s^3}$	$\frac{2}{(s-a)^3}$

What Did We Learn?

Multiplying $f(t)$ by e^{at} in the time domain is equivalent to replacing s with $s-a$ in the Laplace transform. Positive a shifts the pole to the *right*; negative a shifts it to the *left*. No integration required.

5 Property 2 — Second Shifting Theorem (t-Shifting)

Theorem: Second Shifting Theorem

Unit Step Function.

$$u(t-a) = H(t-a) = \begin{cases} 0 & t < a, \\ 1 & t \geq a. \end{cases} \qquad \mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}, \quad a \geq 0.$$

Statement. If $\mathcal{L}\{f(t)\} = F(s)$, then for $a \geq 0$,

$$\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}\,F(s).$$

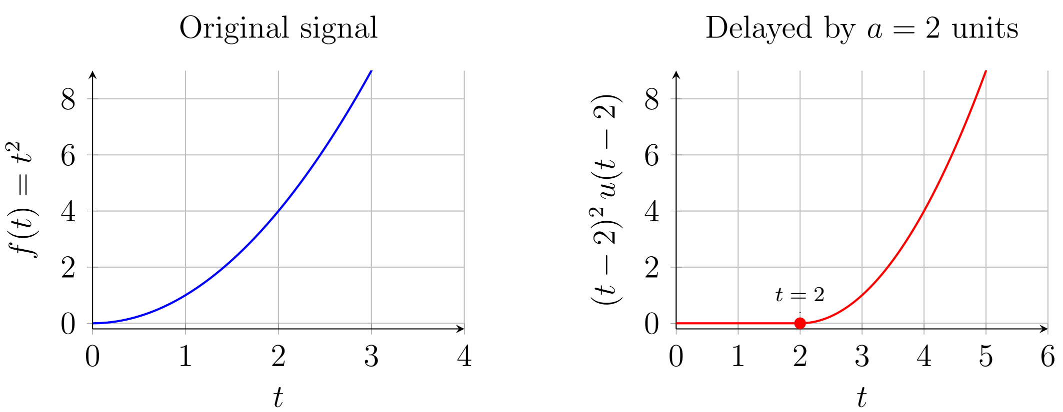
Proof.

$$\begin{aligned} \mathcal{L}\{f(t-a)\,u(t-a)\} &= \int_0^\infty e^{-st}\,f(t-a)\,u(t-a)\,dt \\ &= \int_a^\infty e^{-st}\,f(t-a)\,dt \qquad (\text{since } u(t-a)=0 \text{ for } t<a) \\ &\overset{\tau=t-a}{=} \int_0^\infty e^{-s(\tau+a)}\,f(\tau)\,d\tau \qquad (\text{substitute } \tau = t-a,\ d\tau = dt) \\ &= e^{-as}\int_0^\infty e^{-s\tau}\,f(\tau)\,d\tau \\ &= e^{-as}\,F(s). \end{aligned}$$

Inverse form. $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)\,u(t-a)$.

Visualising the Time Delay

The figure below shows $f(t) = t^2$ (left panel) and its shifted version $f(t-2)\,u(t-2) = (t-2)^2\,u(t-2)$ (right panel). Notice how the parabola is zero for $t < 2$ and then resumes its familiar shape starting at $t = 2$.



Worked Example: Second Shifting Theorem

Find $\mathcal{L}\{(t-2)^2\,u(t-2)\}$.

Step 1. Identify: $f(t) = t^2$ and $a = 2$, so $f(t-2) = (t-2)^2$.

Step 2. Look up the standard table: $\mathcal{L}\{t^2\} = \dfrac{2!}{s^3} = \dfrac{2}{s^3} = F(s)$.
Step 3. Apply the Second Shifting Theorem:

$$\mathcal{L}\{(t-2)^2\, u(t-2)\} = e^{-2s}\, F(s) = \frac{2\, e^{-2s}}{s^3}.$$

Answer: $\dfrac{2\, e^{-2s}}{s^3}$.

What Did We Learn?

Delaying a signal by a in the time domain multiplies its Laplace transform by e^{-as}. Always include the $u(t-a)$ factor — without it, the signal would not be zero before $t = a$, and the Second Shifting Theorem would not apply.

6 Property 3 — Change of Scale

Theorem: Change of Scale Property

Statement. If $\mathcal{L}\{f(t)\} = F(s)$, then for any constant $a > 0$,

$$\mathcal{L}\{f(at)\} = \frac{1}{a}\,F\!\left(\frac{s}{a}\right).$$

Proof.

$$\mathcal{L}\{f(at)\} = \int_0^\infty e^{-st}\,f(at)\,dt.$$

Substitute $u = at$, so $t = u/a$ and $dt = du/a$:

$$= \int_0^\infty e^{-s(u/a)}\,f(u)\,\frac{du}{a} = \frac{1}{a}\int_0^\infty e^{-(s/a)\,u}\,f(u)\,du = \frac{1}{a}\,F\!\left(\frac{s}{a}\right). \qquad \checkmark$$

Worked example: Given $\mathcal{L}\{\sin t\} = \dfrac{1}{s^2+1}$, find $\mathcal{L}\{\sin(3t)\}$ using the Change of Scale property.

Step 1. Here $f(t) = \sin t$ with $F(s) = \dfrac{1}{s^2+1}$, and $a = 3$.
Step 2. Apply the property:

$$\mathcal{L}\{\sin(3t)\} = \frac{1}{3}\,F\!\left(\frac{s}{3}\right) = \frac{1}{3}\cdot\frac{1}{\left(\frac{s}{3}\right)^2 + 1}.$$

Step 3. Simplify the denominator:

$$\left(\frac{s}{3}\right)^2 + 1 = \frac{s^2}{9} + 1 = \frac{s^2+9}{9}.$$

Step 4. Substitute back:

$$\mathcal{L}\{\sin(3t)\} = \frac{1}{3}\cdot\frac{9}{s^2+9} = \frac{3}{s^2+9}. \quad \checkmark$$

This agrees exactly with the standard table result $\mathcal{L}\{\sin(bt)\} = b/(s^2+b^2)$ at $b = 3$.

What Did We Learn?

Scaling time by a (compressing or stretching the signal) scales the s-variable by $1/a$ and introduces an overall factor of $1/a$ in the transform. A larger a (faster signal) moves the transform energy to larger s-values.

7 Property 4 — Multiplication by t^n (Introduction)

Property: Multiplication by t^n

Statement. If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\,F(s), \quad n = 1, 2, 3, \ldots$$

For $n = 1$: $\mathcal{L}\{t\,f(t)\} = -\,F'(s) = -\dfrac{d}{ds}F(s)$.
(The full derivation appears in Topic 10. Here we use the result as a tool.)

Worked example: Find $\mathcal{L}\{t\sin(at)\}$.
Step 1. Let $f(t) = \sin(at)$, so $F(s) = \dfrac{a}{s^2+a^2}$.
Step 2. Apply the $n = 1$ formula:

$$\mathcal{L}\{t\sin(at)\} = -\frac{d}{ds}\left[\frac{a}{s^2+a^2}\right].$$

Step 3. Differentiate using the quotient rule. Let $u = a$ (constant) and $v = s^2 + a^2$. Then $u' = 0$, $v' = 2s$:

$$\frac{d}{ds}\left[\frac{a}{s^2+a^2}\right] = \frac{0\cdot(s^2+a^2) - a\cdot 2s}{(s^2+a^2)^2} = \frac{-2as}{(s^2+a^2)^2}.$$

Step 4. Multiply by -1:

$$\mathcal{L}\{t\sin(at)\} = -\left(\frac{-2as}{(s^2+a^2)^2}\right) = \frac{2as}{(s^2+a^2)^2}.$$

What Did We Learn?

Multiplying $f(t)$ by t corresponds to differentiating $F(s)$ and changing the sign. This avoids computing $\int_0^\infty t\,f(t)\,e^{-st}\,dt$ from scratch. For t^2, differentiate twice; for t^3, differentiate three times.

8 Property 5 — Division by t (Introduction)

Property: Division by t

Statement. If $\mathcal{L}\{f(t)\} = F(s)$ and $\displaystyle\lim_{t\to 0^+}\frac{f(t)}{t}$ exists (is finite), then

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du.$$

(The full derivation is in Topic 11.)

Worked example: Given $\mathcal{L}\{\sin t\} = \dfrac{1}{s^2+1}$, find $\mathcal{L}\left\{\dfrac{\sin t}{t}\right\}$.

Step 1. Check the condition: $\displaystyle\lim_{t\to 0^+}\frac{\sin t}{t} = 1$. The limit is finite, so the property applies.
Step 2. Identify $F(u) = \dfrac{1}{u^2+1}$. Apply the division formula:

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{du}{u^2+1}.$$

Step 3. Evaluate the integral. Recall $\int \frac{du}{u^2+1} = \arctan(u) + C$:

$$\int_s^\infty \frac{du}{u^2+1} = \Big[\arctan(u)\Big]_s^\infty = \frac{\pi}{2} - \arctan(s).$$

Answer: $\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \frac{\pi}{2} - \arctan(s)$.

What Did We Learn?

Dividing $f(t)$ by t corresponds to integrating $F(s)$ from s to ∞. Always verify the existence condition first — if $f(0^+) \neq 0$, the limit $f(t)/t$ diverges and the property cannot be used directly.

9 Summary Properties Table

The following table collects all five properties introduced in this topic, plus two additional standard properties for completeness.

Property Name	Formula	Notes / Validity
Linearity	$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s)$	Always valid
First Shifting (s-shift)	$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$	$s > s_0 + a$
Second Shifting (t-shift)	$\mathcal{L}\{f(t-a)\, u(t-a)\} = e^{-as} F(s)$	$a \geq 0$
Change of Scale	$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$	$a > 0$
Multiplication by t^n	$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$	$n = 1, 2, 3, \ldots$
Division by t	$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\, du$	$\lim_{t\to 0^+} \frac{f(t)}{t}$ must exist
Transform of Derivative	$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$	Requires f piecewise smooth
Transform of Integral	$\mathcal{L}\left\{\int_0^t f(\tau)\, d\tau\right\} = \frac{F(s)}{s}$	Valid for $s > 0$

10 Worked Examples

Example 1 — First Shifting Theorem

Example 1: Find $\mathcal{L}\{e^{-2t}\cos(3t)\}$

Strategy: Use the First Shifting Theorem with $a = -2$ and $f(t) = \cos(3t)$.
Step 1. Look up the standard table:

$$\mathcal{L}\{\cos(3t)\} = \frac{s}{s^2+9} = F(s).$$

Step 2. Apply First Shifting Theorem — replace every s by $(s-(-2)) = (s+2)$:

$$\mathcal{L}\{e^{-2t}\cos(3t)\} = F(s+2) = \frac{s+2}{(s+2)^2+9}.$$

Step 3. Expand the denominator for clarity:

$$(s+2)^2 + 9 = s^2 + 4s + 4 + 9 = s^2 + 4s + 13.$$

Final Answer:

$$\mathcal{L}\{e^{-2t}\cos(3t)\} = \frac{s+2}{s^2+4s+13}.$$

What Did We Learn?

When a is *negative* (decay, e.g., e^{-2t}), the shift goes to the *left* in the complex s-plane (replace s by $s+|a|$). The denominator becomes a completed square, which is the fingerprint of cosine (or sine) shifted.

Example 2 — Second Shifting Theorem with Piecewise Function

Example 2: Piecewise function via unit step

Given:

$$f(t) = \begin{cases} 0 & 0 \le t < 1, \\ t-1 & t \ge 1. \end{cases}$$

Step 1. Express using unit step function. The function switches from 0 to $(t-1)$ at $t = 1$. We write:

$$f(t) = (t-1)\,u(t-1).$$

Verification: for $t < 1$, $u(t-1) = 0$ so $f(t) = 0$. For $t \ge 1$, $u(t-1) = 1$ so $f(t) = t - 1$. ✓
Step 2. Identify the shifted function. Here $g(t) = t$, shifted by $a = 1$: $g(t-1) = t-1$.
From the table: $\mathcal{L}\{g(t)\} = \mathcal{L}\{t\} = \dfrac{1}{s^2} = G(s)$.
Step 3. Apply the Second Shifting Theorem.

$$\mathcal{L}\{(t-1)\,u(t-1)\} = e^{-s}\,G(s) = \frac{e^{-s}}{s^2}.$$

Final Answer: $\mathcal{L}\{f(t)\} = \dfrac{e^{-s}}{s^2}$.

What Did We Learn?

Any piecewise function that "turns on" at $t = a$ can be written as $g(t-a)\,u(t-a)$ for an appropriate g. Once in that form, the Second Shifting Theorem gives the Laplace transform in one step.

Example 3 — First Shifting Theorem applied to t^2

Example 3: Find $\mathcal{L}\{t^2 e^{-t}\}$

Strategy: Use the First Shifting Theorem with $a = -1$ and $f(t) = t^2$.
Step 1. Standard table entry:

$$\mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3} = F(s).$$

Step 2. Apply First Shifting Theorem with $a = -1$ (i.e., replace s by $s-(-1) = s+1$):

$$\mathcal{L}\{t^2 e^{-t}\} = F(s+1) = \frac{2}{(s+1)^3}.$$

Step 3. No further simplification is needed; the answer is already compact.
Final Answer:

$$\mathcal{L}\{t^2 e^{-t}\} = \frac{2}{(s+1)^3}.$$

What Did We Learn?

For $e^{at}t^n$, the Laplace transform is always $n!/(s-a)^{n+1}$. The First Shifting Theorem makes this a one-step substitution from the t^n table entry, regardless of n.

11 Excel Example

Objective: Numerically verify the First Shifting Theorem for $\mathcal{L}\{e^{2t}\sin(t)\}$ at $s = 4$.
The theorem predicts:

$$\mathcal{L}\{e^{2t}\sin(t)\}\Big|_{s=4} = \frac{1}{(4-2)^2+1} = \frac{1}{4+1} = \frac{1}{5} = 0.2.$$

We approximate the Laplace integral $\int_0^\infty e^{-4t}\,e^{2t}\sin(t)\,dt = \int_0^\infty e^{-2t}\sin(t)\,dt$ numerically using a Riemann sum with step $\Delta t = 0.1$.

Col A t	Col B Integrand $e^{-4t}e^{2t}\sin t$	Col C Approx. (B×0.1)	Col D Running Sum	Col E Exact (0.2)
0.0	`=EXP(-4*A2)*EXP(2*A2)*SIN(A2)`	`=B2*0.1`	`=C2`	0.2
0.1	`=EXP(-4*A3)*EXP(2*A3)*SIN(A3)`	`=B3*0.1`	`=D2+C3`	0.2
0.2	`=EXP(-4*A4)*EXP(2*A4)*SIN(A4)`	`=B4*0.1`	`=D3+C4`	0.2
0.3	`=EXP(-4*A5)*EXP(2*A5)*SIN(A5)`	`=B5*0.1`	`=D4+C5`	0.2
0.4	`=EXP(-4*A6)*EXP(2*A6)*SIN(A6)`	`=B6*0.1`	`=D5+C6`	0.2
⋮	⋮	⋮	⋮	0.2
6.0	`=EXP(-4*A62)*EXP(2*A62)*SIN(A62)`	`=B62*0.1`	`=D61+C62`	0.2

How to set up the spreadsheet:

1. In column A, enter $t = 0.0$ in cell A2, then $t = 0.1$ in A3. Select A2:A3 and drag down to row 62 (covering $t = 0.0$ to $t = 6.0$).
2. In column B (cell B2), enter: `=EXP(-4*A2)*EXP(2*A2)*SIN(A2)` and drag down. This computes the integrand $e^{-4t} \cdot e^{2t} \cdot \sin(t) = e^{-2t}\sin(t)$.
3. In column C (cell C2), enter `=B2*0.1` (rectangle width $\Delta t = 0.1$) and drag down. This is each rectangle's area.
4. In column D (cell D2), enter `=C2`; in D3 enter `=D2+C3`. Drag D3 down to row 62. Cell D62 is the running Riemann sum — the numerical approximation.
5. In column E, place the exact value 0.2 in every row for comparison.

Expected result: The running sum in column D should converge to approximately 0.200 as t increases, confirming that $1/((4-2)^2+1) = 0.2$.

What Did We Learn?

The numerical integration confirms the First Shifting Theorem: $e^{-4t} \cdot e^{2t}\sin(t) = e^{-2t}\sin(t)$, and the integral equals $1/5 = 0.2$, matching $1/((s-a)^2+b^2)$ at $s = 4$, $a = 2$, $b = 1$. Numerical verification builds confidence in the algebraic theorem.

12 Python Example

Listing 1: Verifying Laplace Transform Properties using SymPy

```python
import sympy as sp

t, s, a, b, u = sp.symbols('t s a b u', positive=True)

# ------------------------------------------------------
# 1. FIRST SHIFTING THEOREM
#    L{e^{at} f(t)} should equal F(s-a)
# ------------------------------------------------------

def verify_first_shift(f_expr, label):
    """Compute L{e^{at}*f(t)} directly and compare to F(s-a)."""
```

```
    # Direct Laplace of e^{at}*f(t)
    direct = sp.laplace_transform(sp.exp(a*t) * f_expr, t, s, noconds=True)
    # Laplace of f(t) then substitute s -> s-a
    F_s     = sp.laplace_transform(f_expr, t, s, noconds=True)
    shifted = F_s.subs(s, s - a)
    match   = sp.simplify(direct - shifted) == 0
    print(f"{label}: direct={direct}, F(s-a)={shifted}, match={match}")

verify_first_shift(sp.cos(b*t), "f(t)=cos(bt)")
# Output: f(t)=cos(bt): direct=(s-a)/((s-a)^2+b^2), F(s-a)=(s-a)/((s-a)^2+b^2), match=True

verify_first_shift(t**2, "f(t)=t^2")
# Output: f(t)=t^2: direct=2/(s-a)^3, F(s-a)=2/(s-a)^3, match=True

verify_first_shift(sp.sin(b*t), "f(t)=sin(bt)")
# Output: f(t)=sin(bt): direct=b/((s-a)^2+b^2), F(s-a)=b/((s-a)^2+b^2), match=True

print()

# -------------------------------------------------------
# 2. CHANGE OF SCALE PROPERTY
#    L{f(at)} should equal (1/a)*F(s/a)
# -------------------------------------------------------

f_expr_scale = sp.exp(-t)   # f(t) = e^{-t}, so f(at) = e^{-at}
F_s_scale    = sp.laplace_transform(f_expr_scale, t, s, noconds=True)
# F(s) = 1/(s+1), so F(s/a) = 1/(s/a + 1) = a/(s+a)

# Direct: L{e^{-at}} = 1/(s+a)
direct_scale = sp.laplace_transform(sp.exp(-a*t), t, s, noconds=True)
# Property: (1/a)*F(s/a)
property_scale = sp.Rational(1,1)/a * F_s_scale.subs(s, s/a)

print("Change of Scale for f(t)=e^{-t}:")
print(f"  Direct L{{e^{{-at}}}}  = {sp.simplify(direct_scale)}")
# Output: Direct L{e^{-at}}  = 1/(a+s)
print(f"  (1/a)*F(s/a)      = {sp.simplify(property_scale)}")
# Output: (1/a)*F(s/a)      = 1/(a+s)
print(f"  Match: {sp.simplify(direct_scale - property_scale) == 0}")
# Output: Match: True
```

What Did We Learn?

SymPy's `laplace_transform` function confirms all three tests of the First Shifting Theorem (for $\cos(bt)$, t^2, $\sin(bt)$) and the Change of Scale property (for e^{-t}). Symbolic verification is an excellent way to cross-check hand calculations before an exam.

13 Viva-Style Oral Questions

1. **State the First Shifting Theorem.**
Answer. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ for $s > s_0+a$. Multiplying a time-domain signal by e^{at} is equivalent to shifting the transform variable from s to $s-a$.

2. **Outline the proof of the First Shifting Theorem.**
Answer. Write the Laplace integral: $\mathcal{L}\{e^{at}f(t)\} = \int_0^\infty e^{-st}e^{at}f(t)\,dt = \int_0^\infty e^{-(s-a)t}f(t)\,dt = F(s-a)$. The key step is combining the two exponentials into one, revealing that s is simply replaced by $s-a$ in the original transform.

3. **State the Second Shifting Theorem.**
Answer. If $\mathcal{L}\{f(t)\} = F(s)$ and $a \geq 0$, then $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}F(s)$. Delaying a signal by a in the time domain multiplies the Laplace transform by e^{-as}.

4. **What does the unit step function $u(t-a)$ do in the Second Shifting Theorem?**
Answer. It "switches on" the delayed signal $f(t-a)$ at $t=a$ and keeps it at zero for all $t < a$. Without $u(t-a)$, the function $f(t-a)$ would exist for $t < a$ (possibly with negative t arguments), which violates causality and makes the theorem inapplicable.

5. **Why does the factor e^{-as} appear in the Second Shifting Theorem?**
Answer. It comes from the substitution $\tau = t-a$ in the integral $\int_a^\infty e^{-st}f(t-a)\,dt$. After substituting, $t = \tau + a$, so $e^{-st} = e^{-s(\tau+a)} = e^{-sa}\,e^{-s\tau}$. The factor e^{-sa} (constant w.r.t. τ) comes outside the integral, yielding $e^{-as}F(s)$.

6. **State the Change of Scale property and give the formula.**
Answer. If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$, then $\mathcal{L}\{f(at)\} = \frac{1}{a}F(s/a)$. Compressing time by a factor a stretches the transform by a (argument s becomes s/a) and reduces the overall amplitude by $1/a$.

7. **State the Multiplication by t property.**
Answer. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{t\,f(t)\} = -F'(s)$. More generally, $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$. Each multiplication by t corresponds to one differentiation of the transform and a sign change.

8. **When is the Division by t property valid?**
Answer. The property $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ holds if and only if $\lim\limits_{t\to 0^+} \dfrac{f(t)}{t}$ is finite. For example, $f(t) = \sin t$ satisfies this since $\sin(t)/t \to 1$ as $t \to 0^+$. But $f(t) = t^{-1/2}$ does not, because $t^{-1/2}/t = t^{-3/2} \to \infty$.

14 Descriptive Questions

1. **State and prove the First Shifting Theorem.**
State: If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$. Proof: $\int_0^\infty e^{-st}e^{at}f(t)\,dt = \int_0^\infty e^{-(s-a)t}f(t)\,dt = F(s-a)$. Explain the domain constraint $s > s_0 + a$ and give two applications from the standard table.

2. **State and prove the Second Shifting Theorem.**
State: If $\mathcal{L}\{f(t)\} = F(s)$ and $a \geq 0$, then $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}F(s)$. Proof: Split the integral at $t = a$, substitute $\tau = t - a$, and factor out e^{-as}. Explain the role of the unit step function.

3. **Derive the Change of Scale property.**
Starting from the definition $\mathcal{L}\{f(at)\} = \int_0^\infty e^{-st}f(at)\,dt$, substitute $u = at$ to obtain $\frac{1}{a}F(s/a)$. Show every algebraic step and verify the result using $f(t) = e^{-t}$ (so $f(at) = e^{-at}$ and $F(s) = 1/(s+1)$).

4. **Find $\mathcal{L}\{te^{at}\}$ using the Multiplication by t property.**
Let $f(t) = e^{at}$ with $\mathcal{L}\{e^{at}\} = 1/(s-a) = F(s)$. By the property: $\mathcal{L}\{t \cdot e^{at}\} = -F'(s) = -\frac{d}{ds}\left[\frac{1}{s-a}\right] = \frac{1}{(s-a)^2}$. Show the differentiation in full. Confirm for $a = 0$: $\mathcal{L}\{t\} = 1/s^2$. ✓

5. **Find $\mathcal{L}\{(1-\cos t)/t\}$ using the Division by t property.**
Let $f(t) = 1-\cos t$. Check: $\lim_{t\to 0^+}(1-\cos t)/t = 0$ (finite). So the property applies. $F(s) = \mathcal{L}\{1-\cos t\} = 1/s - s/(s^2+1)$. Then $\mathcal{L}\{(1-\cos t)/t\} = \int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2+1}\right) du = \left[\ln u - \frac{1}{2}\ln(u^2+1)\right]_s^\infty = \frac{1}{2}\ln\left(1 + \frac{1}{s^2}\right)$.

15 Practice Problems

1. Find $\mathcal{L}\{e^{-3t}t^2\}$.
Hint: Use First Shifting Theorem with $f(t) = t^2$, $a = -3$. *Answer:* $2/(s+3)^3$.

2. Find $\mathcal{L}\{e^t\cos(2t)\}$.
Hint: $\mathcal{L}\{\cos(2t)\} = s/(s^2+4)$; shift $s \to s-1$. *Answer:* $(s-1)/\big((s-1)^2+4\big)$.

3. Find $\mathcal{L}\{(t-1)\,u(t-1)\}$.
Hint: Second Shifting Theorem with $f(t) = t$, $a = 1$. *Answer:* e^{-s}/s^2.

4. Find $\mathcal{L}\{t\cos(at)\}$ using the Multiplication by t property.
Hint: $F(s) = s/(s^2+a^2)$; compute $-F'(s)$. *Answer:* $(s^2-a^2)/(s^2+a^2)^2$.

5. Find $\mathcal{L}\{(e^{at}-1)/t\}$ using the Division by t property.
Hint: $\mathcal{L}\{e^{at}-1\} = 1/(s-a) - 1/s$; integrate from s to ∞. *Answer:* $\ln\big(s/(s-a)\big)$ (valid for $s > a > 0$).

6. Find $\mathcal{L}\{\sin(2t+\pi/3)\}$.
Hint: Expand using the angle-addition formula: $\sin(2t+\pi/3) = \sin(2t)\cos(\pi/3) + \cos(2t)\sin(\pi/3) = \frac{1}{2}\sin(2t) + \frac{\sqrt{3}}{2}\cos(2t)$; then apply linearity and the standard table.
Answer: $\dfrac{1}{2} \cdot \dfrac{2}{s^2+4} + \dfrac{\sqrt{3}}{2} \cdot \dfrac{s}{s^2+4} = \dfrac{2+\sqrt{3}s}{2(s^2+4)}$.

16 MCQs

1. If $\mathcal{L}\{\sin(bt)\} = b/(s^2+b^2)$, then $\mathcal{L}\{e^{at}\sin(bt)\}$ equals:
   (a) $\dfrac{b}{(s+a)^2+b^2}$ **(b)** $\dfrac{b}{(s-a)^2+b^2}$ (c) $\dfrac{b}{s^2+b^2}\,e^{-as}$ (d) $\dfrac{b}{(s^2-a^2)+b^2}$
   Explanation: First Shifting Theorem replaces s by $s-a$, giving option (b).

2. The Laplace transform $\mathcal{L}\{f(t-a)\,u(t-a)\}$ equals:
   (a) $F(s)$ (b) $e^{as}F(s)$ **(c)** $e^{-as}F(s)$ (d) $F(s-a)$
   Explanation: Second Shifting Theorem: delaying by a introduces e^{-as} (negative sign).

3. If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$, then $\mathcal{L}\{f(at)\}$ equals:
   (a) $a\,F(as)$ **(b)** $\dfrac{1}{a}\,F\left(\dfrac{s}{a}\right)$ (c) $F(as)$ (d) $\dfrac{1}{a}\,F(as)$
   Explanation: Change of Scale: both the $1/a$ factor and the s/a argument are required.

4. Using Multiplication by t, $\mathcal{L}\{t\,e^{at}\}$ equals:
   (a) $\dfrac{1}{(s-a)}$ (b) $\dfrac{1}{(s-a)^3}$ **(c)** $\dfrac{1}{(s-a)^2}$ (d) $\dfrac{2}{(s-a)^3}$
   Explanation: $-\dfrac{d}{ds}\left[\dfrac{1}{s-a}\right] = \dfrac{1}{(s-a)^2}$.

5. In the Second Shifting Theorem $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}F(s)$, the factor e^{-as} arises because:
   (a) e^{-as} is the Laplace transform of e^{-at} (b) the theorem uses complex frequency $s+a$ **(c) substituting $\tau = t-a$ introduces e^{-as} as a constant factor outside the integral** (d) the unit step function contributes e^{-as} independently
   Explanation: After substitution $\tau = t-a$, the term $e^{-s(\tau+a)} = e^{-as}e^{-s\tau}$; the e^{-as} factor is constant and comes out of the integral, giving $e^{-as}F(s)$.

17 Common Mistakes

Common Mistakes in Properties of Laplace Transforms

Mistake	What Students Do	Correct Approach
Shifting t instead of s	Write $\mathcal{L}\{e^{at}f(t)\} = e^{-as}F(s)$ (confusing with Second Shifting)	First Shifting: replace s by $s-a$ in $F(s)$; never multiply by e^{-as}
Forgetting $u(t-a)$ in the inverse	Write $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)$ (missing the unit step)	Always write $f(t-a)\,u(t-a)$; the step function is part of the answer
Change of Scale without $1/a$ factor	Write $\mathcal{L}\{f(at)\} = F(s/a)$ (omitting the $1/a$ prefactor)	The correct formula is $\mathcal{L}\{f(at)\} = \frac{1}{a}F(s/a)$
Using Division by t without checking limit	Apply $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ when $f(0^+)/0$ is undefined	First verify $\lim_{t\to 0^+} f(t)/t$ exists; if not, the property cannot be applied
Sign error in First Shifting	Use $F(s+a)$ instead of $F(s-a)$ when $a > 0$	Multiplying by e^{at} shifts *left* by a in s (replace s by $s-a$, not $s+a$)

18 Quick Recap

Quick Recap — Properties of Laplace Transforms

- **First Shifting (s-shift):** $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ — replace s with $s-a$.
- **Second Shifting (t-shift):** $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}F(s)$ — always include $u(t-a)$.
- **Change of Scale:** $\mathcal{L}\{f(at)\} = \frac{1}{a}F(s/a)$ — both the factor $1/a$ and the argument s/a are required.
- **Multiplication by t^n:** $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$ — differentiate $F(s)$ n times and multiply by $(-1)^n$.
- **Division by t:** $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ — only valid when $\lim_{t\to 0^+} f(t)/t$ is finite.
- **s-shift vs. t-shift:** s-shifting multiplies by e^{at} in time; t-shifting multiplies by e^{-as} in frequency.
- **Unit step function key:** $u(t-a)$ is *always* part of the answer for Second Shifting — forgetting it is the most common exam error.
- **Use the summary table:** Keep the 8-row properties table handy during problem solving — it covers all scenarios in this topic.